RESEARCH STATEMENT

SITRAKA RANDRIANARIVO

Koetter and Kschischang introduced the operator-channel framework for communication through subspaces over finite fields, modeling rank loss as erasures and interference as errors [3]. Soleymani and Mahdavifar extended this framework to complex vector spaces through the analog operator channel, motivated by non-coherent wireless networks [4]. Information is encoded in a complex vector subspace: rank loss can reduce the transmitted subspace, while interference can add unwanted directions. The receiver must recover the transmitted subspace from this altered observation. My primary research is in algebraic methods for coding theory, especially the construction and analysis of subspace codes for the analog operator channel.

My current work with Felice Manganiello concerns two algebraic approaches to coding in this setting. One project studies character-polynomial codes, including their exact cardinality and parametrization. The other constructs complex subspace codes from finite-field subspace codes using extended multiplicative characters. Together, these projects examine how algebraic constructions and character maps affect code size, distance, and the relationship between messages and codewords.

Character-Polynomial Codes

Character-polynomial (CP) codes were introduced by Soleymani and Mahdavifar as an algebraic construction for analog subspace coding [4]. They are one-dimensional complex subspace codes formed by evaluating finite-field polynomials and mapping their values to complex roots of unity using an additive character.

In our paper, *Exact Cardinality and Nonredundant Parametrization of Character-Polynomial Codes* [5], we identify redundancy in the usual polynomial description over extension fields: distinct polynomials can generate the same codeword. This redundancy arises because an additive character of an extension field factors through the field trace to its prime field, and therefore may not distinguish all finite-field evaluation values.

We analyze the trace-evaluation map using p-cyclotomic cosets and determine the exact number of distinct codewords. In the paper's standard evaluation setting, the cardinality is

$$|\mathcal{C}(\mathcal{F})| = p^{\sum_{\ell \in L} |C_p(\ell)| - 1},$$

where L is a set of representatives for the relevant p-cyclotomic cosets and $C_p(\ell)$ is the coset represented by ℓ. We also construct a refined set of defining polynomials whose map to the code is a bijection: every codeword has exactly one representative in this set. These results give an exact code-size count and a nonredundant polynomial parametrization, clarifying how polynomial messages correspond to codewords in rate calculations and future coding and decoding work.

Complex Subspace Codes from Finite Fields

In a second project, we construct complex subspace codes from finite-field subspace codes using an extended bijective multiplicative character [6]. The analog operator channel motivates this

construction: it allows us to bring established algebraic families over finite fields into the complex setting while controlling their code parameters.

First, we establish an exact relation between the subspace metric and the squared Frobenius distance between orthogonal projection matrices, the distance used in the analog-channel framework of Soleymani and Mahdavifar [4]. Within their error-and-erasure correction framework, we show that the subspace metric provides a guarantee at least as strong as the one based on squared projection distance, and can provide a strictly stronger guarantee. This comparison motivates our study of complex subspaces equipped with the subspace metric as a setting for error-correcting codes for the analog operator channel.

To construct codes, we fix a basis matrix for each finite-field codeword, apply the extended character entrywise, and take the complex row space of the resulting matrix. Since the character map is not additive, different basis choices can produce different complex realizations. For every choice, however, the construction preserves cardinality and codeword dimensions and does not decrease the minimum subspace distance.

We prove a stronger distance result for character image codes of lifted maximum-rank-distance (MRD) codes. Let $\mathcal{C}_R \subseteq \mathbb{F}_q^{l\times m}$ be an MRD code. Its lift consists of the subspaces $\mathrm{rs}_{\mathbb{F}_q}\begin{pmatrix} I_l & M\end{pmatrix}$, where $M \in \mathcal{C}_R$. Using these systematic basis matrices, the character image code of the lift is the lift of the character image code of $\mathcal{C}_R$:

$$\chi^*\big(\mathrm{Lift}(\mathcal{C}_R)\big) = \mathrm{Lift}\big(\chi(\mathcal{C}_R)\big).$$

We prove that the character image code of a lifted MRD code has the same cardinality and minimum subspace distance as the original finite-field lifted MRD code.

The cardinality equality follows from the general character construction. For the distance result, we study $\chi(\mathcal{C}_R)$ as a rank-metric code with entries in $\boldsymbol{\mu}_{q-1}^0$, the alphabet consisting of zero and the relevant roots of unity, with rank computed over $\mathbb{C}$. A Singleton-type bound for these rank-metric codes, combined with the MRD cardinality and the rank comparison between $\mathcal{C}_R$ and $\chi(\mathcal{C}_R)$, forces equality of their minimum rank distances. The lifting identity $d_S = 2d_R$ in [8] then gives equality of minimum subspace distances.

We also derive a Singleton-type bound for complex subspace codes in the fixed systematic lifting chart. Specifically, consider codes whose codewords have the form $\mathrm{rs}_{\mathbb{C}}\begin{pmatrix} I_l & M\end{pmatrix}$, where $M \in (\boldsymbol{\mu}_{q-1}^0)^{l\times m}$. For such a code $\mathcal{C}$, the bound is

$$|\mathcal{C}| \le |\boldsymbol{\mu}_{q-1}^0|^{\,m\left(l-\frac{d_S(\mathcal{C})}{2}+1\right)}.$$

We prove that character image codes of lifted MRD codes attain this bound, so they have maximum cardinality within this specified family for their minimum subspace distance. This result applies to lifted MRD codes generally. The Koetter–Kschischang code is one example, since it is obtained by lifting a Gabidulin code [3, 7].

Future Directions

Error-Free Decoding of Character-Image Lifted Gabidulin Codes. A more specific problem, currently under investigation, is error-free decoding for character images of lifted Gabidulin codes. Suppose the received subspace $U = \mathrm{rs}_{\mathbb{C}}(Y)$ is contained in a codeword V_D, where $D \in \chi(\mathcal{C}_R)$ and $Y = [\,A \mid B\,]$. Then $B = AD$. Since A has full row rank N, each column d_j of D must satisfy $Ad_j = b_j$, where b_j is the corresponding column of B. The solutions to each such system form an affine $\mathbb{C}$-space of dimension $\rho = \ell - N$. The first challenge is to find the points in these affine spaces whose coordinates belong to the finite character alphabet $\boldsymbol{\mu}_{q-1}^0$. A further challenge is to exploit the

structure of the character image of the Gabidulin code to narrow these candidates. This requires understanding how polynomial evaluations, the chosen basis for representing finite-field elements, and the bijective multiplicative character constrain the possible matrices D. Our goal is to use these constraints to develop an efficient decoding algorithm.

Decoding Character-Image Lifted MRD Codes. A natural direction for future work is to develop efficient decoding algorithms for character-image codes of lifted MRD codes. Let $\mathcal{C}_R \subseteq \mathbb{F}_q^{\ell \times m}$ be a rank-metric code, and write $\chi(\mathcal{C}_R)$ for its entrywise character image. For each $D \in \chi(\mathcal{C}_R)$, define

$$V_D = \mathrm{rs}_{\mathbb{C}}([\, I_\ell \mid D \,]).$$

Suppose the received subspace is $U = \mathrm{rs}_{\mathbb{C}}(Y)$, where $Y = [\, A \mid B \,] \in \mathbb{C}^{N \times (\ell+m)}$ has full row rank. For a row vector $c \in \mathbb{C}^{1 \times N}$, the vector $cY = (cA, cB)$ belongs to V_D if and only if $cB = (cA)D$, or equivalently $c(B - AD) = 0$. Since Y has full row rank, the map $c \mapsto cY$ is injective. It follows that

$$\dim(U \cap V_D) = N - \mathrm{rank}_{\mathbb{C}}(B - AD).$$

Since $\dim(U) = N$ and $\dim(V_D) = \ell$, the subspace distance is

$$d_S(U, V_D) = \ell - N + 2\,\mathrm{rank}_{\mathbb{C}}(B - AD).$$

The term $\ell - N$ is independent of D. Thus, minimum-distance decoding reduces to finding

$$\widehat{D} \in \operatorname*{arg\,min}_{D \in \chi(\mathcal{C}_R)} \mathrm{rank}_{\mathbb{C}}(B - AD).$$

This gives a minimum-distance decoding criterion, but an efficient algorithm for solving the minimization remains open.

A related row-reduction approach was developed by Silva, Kschischang, and Kötter [8] for lifted rank-metric codes over $\mathbb{F}_q$. Motivated by their generalized rank-decoding formulation, one can reduce the received matrix Y over $\mathbb{C}$ and express the decoding objective for $\chi(\mathcal{C}_R)$ as

$$\widehat{D} \in \operatorname*{arg\,min}_{D \in \chi(\mathcal{C}_R)} \mathrm{rank}_{\mathbb{C}} \begin{bmatrix} \widehat{L} & R - D \\ 0 & \widehat{E} \end{bmatrix}.$$

Here, R is the right-hand block obtained after row reduction, while $\widehat{L}$ and $\widehat{E}$ encode additional information about the received subspace. The rank of this block matrix differs from $\mathrm{rank}_{\mathbb{C}}(B - AD)$ by a constant independent of D, so the two objectives have the same minimizers. This reformulation may help identify structure that an efficient decoder for $\chi(\mathcal{C}_R)$ could exploit. However, the algorithms in [8] do not transfer directly, since they rely on algebraic structure that the nonadditive character map χ does not automatically preserve.

List Decoding via the Plücker Embedding. Another direction is to investigate list decoding for character-image codes of lifted codes through the Plücker embedding. For lifted Gabidulin codes over finite fields, selected Plücker coordinates satisfy parity-check equations that can be combined with equations describing a decoding ball to produce a list of nearby codewords [1]. These parity-check equations do not automatically carry over to the character-image setting: the Plücker coordinates are minors of the matrices $[\, I_\ell \mid \chi(M) \,]$, and the character map does not preserve the additions underlying the finite-field linear constraints. A first step is to characterize the Plücker coordinates of the character-image code and determine how they are constrained by the finite alphabet $\boldsymbol{\mu}_{q-1}^0$. Since a ball in the complex Grassmannian generally contains infinitely many subspaces, the list-decoding problem should focus on its intersection with the finite set of subspaces that admit a basis matrix with entries in $\boldsymbol{\mu}_{q-1}^0$. Characterizing this set and its intersection with a decoding ball may provide a basis for a list-decoding algorithm.

General Bounds for Character-Image Subspace Codes. I also plan to investigate general bounds on the cardinality and minimum distance of character-image subspace codes, extending beyond the fixed systematic lifting chart considered in my current work.

Character-Image Codes under the Squared Projection Frobenius Distance. Another direction is to study character-image codes with respect to the 2-quasi metric

$$d_2(U, V) = \|P_U - P_V\|_F^2,$$

where P_U and P_V are the orthogonal projection matrices onto U and V. Soleymani and Mahdavifar studied the minimum distance of character-polynomial codes under this metric; these codes are obtained by applying an additive character to evaluation vectors from a subset of a Reed–Solomon code [4]. In our setting, a related one-dimensional code can be formed by evaluating a chosen finite family of monic polynomials and applying the extended bijective multiplicative character. The resulting code is in bijection with that polynomial family. I would like to determine its minimum-distance properties under d_2 and, more broadly, understand how character images code of finite-field subspace codes behave under this 2-quasi metric.

Character Images of Other Subspace Codes. I would also like to study character-image constructions from other well-known families of finite-field subspace codes, including orbit codes, and analyze how their parameters and basis choices affect the resulting complex codes.

Long-Term Interests. My broader interests include commutative algebra, algebraic number theory, and computational algebraic geometry, particularly where these subjects connect with algebraic coding theory. These are areas I hope to explore alongside my current work. My long-term goal is to contribute to algebraic coding theory through constructions with precise parameter guarantees and algebraic structure that can support decoding over complex spaces.

References

[1] A.-L. Trautmann, N. Silberstein, and J. Rosenthal, "List decoding of lifted Gabidulin codes via the Plücker embedding," in *Proceedings of the International Workshop on Coding and Cryptography (WCC)*, 2013, pp. 539–549.

[2] D. Silva, F. R. Kschischang, and R. Koetter, "A rank-metric approach to error control in random network coding," *IEEE Transactions on Information Theory*, vol. 54, no. 9, pp. 3951–3967, 2008.

[3] R. Koetter and F. R. Kschischang, "Coding for errors and erasures in random network coding," *IEEE Transactions on Information Theory*, vol. 54, no. 8, pp. 3579–3591, 2008.

[4] M. Soleymani and H. Mahdavifar, "Analog subspace coding: A new approach to coding for non-coherent wireless networks," *IEEE Transactions on Information Theory*, vol. 68, no. 4, pp. 2349–2364, 2022. doi: 10.1109/TIT.2021.3133071.

[5] F. Manganiello and S. Randrianarivo, "Exact cardinality and nonredundant parametrization of character-polynomial codes," arXiv:2607.11595, 2026. `https://arxiv.org/abs/2607.11595`.

[6] F. Manganiello and S. Randrianarivo, "Subspace code construction over the complex numbers using extended multiplicative characters," doi: 10.13140/RG.2.2.33418.43208. Under review.

[7] E. M. Gabidulin, "Theory of codes with maximum rank distance," *Problems of Information Transmission*, vol. 21, no. 1, pp. 1–12, 1985.

[8] D. Silva, F. R. Kschischang, and R. Koetter, "A rank-metric approach to error control in random network coding," *IEEE Transactions on Information Theory*, vol. 54, no. 9, pp. 3951–3967, 2008.